Problems - Week 9

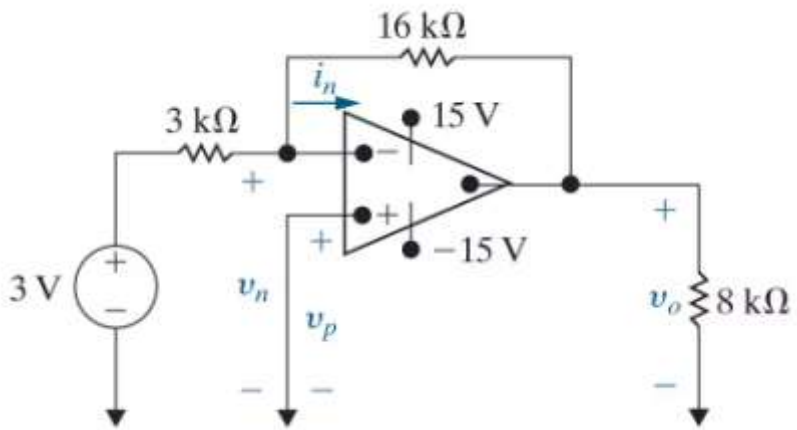
- Operational Amplifier

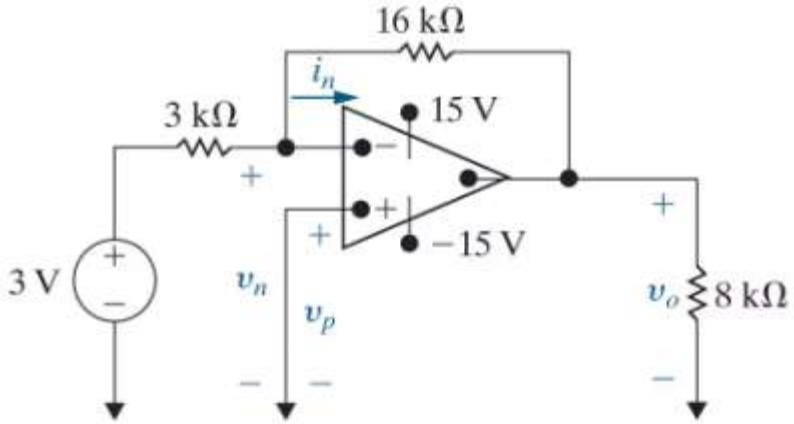
a. Label the 5 op amp terminals with their names.

b. What ideal op amp constraint determines the value of i_n? What is this value?

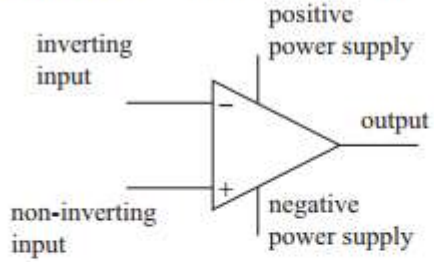
c. What ideal op amp constraint determines the value of $v_p - v_n$? What is this value?

d. Calculate v_o.





[a] The five terminals of the op amp are identified as follows:



[b] The input resistance of an ideal op amp is infinite, which constrains the value of the input currents to 0. Thus, $i_n = 0$ A.

[c] The open-loop voltage gain of an ideal op amp is infinite, which constrains the difference between the voltage at the two input terminals to 0. Thus, $v_p - v_n = 0$.

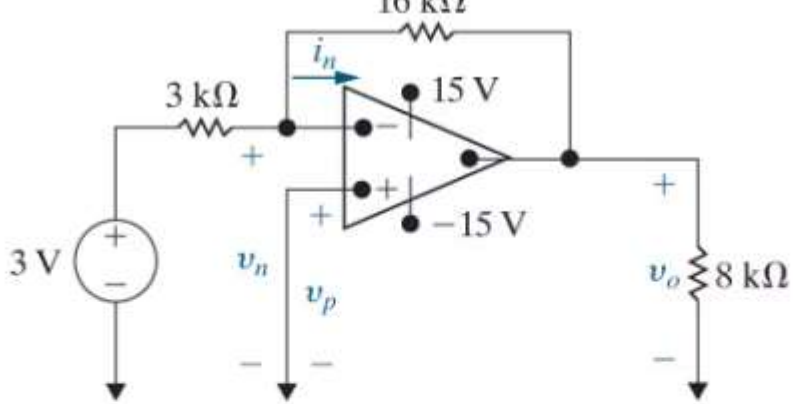
[d] Write a node voltage equation at v_n:

$$\frac{v_n - 3}{3000} + \frac{v_n - v_o}{16{,}000} = 0.$$

But $v_p = 0$ and $v_n = v_p = 0$. Thus,

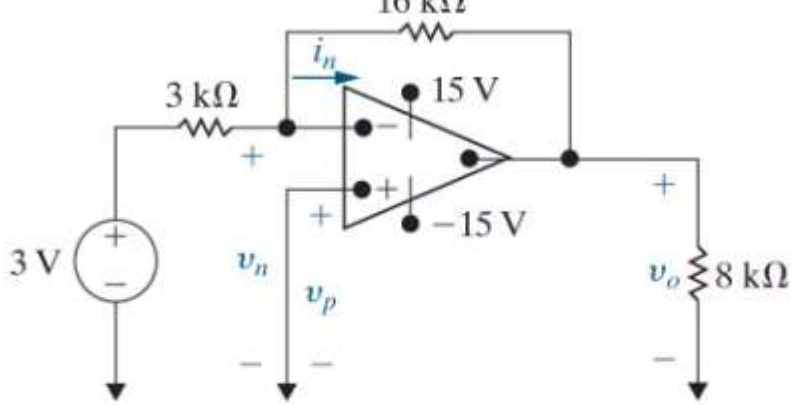
$$\frac{-3}{3000} - \frac{v_o}{16{,}000} = 0 \quad \text{so} \quad v_o = -16 \text{ V}.$$

Note that v_o is lower than the negative supply voltage, so our assumption that the op amp is in its linear operating region is not correct. Thus, v_o saturates at $v_o = -15$ V.



a. Replace the 2.5 V source in the circuit in Fig. P5.1 and calculate v_o for each of the following source values: 0.4V, 2.0V, 3.5V, −0. 6V, −1. 6V, −2 .4V

b. Specify the range of voltage source values that will not cause the op amp to saturate.



a. Replace the 2.5 V source in the circuit in Fig. P5.1 and calculate v_o for each of the following source values: 0.4V, 2.0V, 3.5V, −0. 6V, −1. 6V, −2 .4V

b. Specify the range of voltage source values that will not cause the op amp to saturate.

Let the value of the voltage source be v_s:

$$\frac{v_n - v_s}{3000} + \frac{v_n - v_o}{16{,}000} = 0.$$

But $v_n - v_p = 0$. Therefore,

$$v_o = -\frac{16{,}000}{3000}v_s = -\frac{16}{3}v_s.$$

When $v_s = -5$ V, $v_o = -\frac{16}{3}(-5) = 26.67$ V; saturates at $v_o = 15$ V.
When $v_s = -7$ V, $v_o = -\frac{16}{3}(-7) = 37.33$ V; saturates at $v_o = 15$ V.
When $v_s = 5$ V, $v_o = -\frac{16}{3}(-5) = -26.67$ V; saturates at $v_o = -15$ V.
When $v_s = 7$ V, $v_o = -\frac{16}{3}(7) = -37.33$ V; saturates at $v_o = -15$ V.

[b] $-\frac{16}{3}v_s = 15$ so $v_s = -2.81$ V

$-\frac{16}{3}v_s = -15$ so $v_s = 2.81$ V

The range of source voltages that avoids saturation is $-2.81 \text{ V} \leqslant v_s \leqslant 2.81 \text{ V}$.

a. Calculate v_o if $v_a = 2$ V and $v_b = 1$ V.

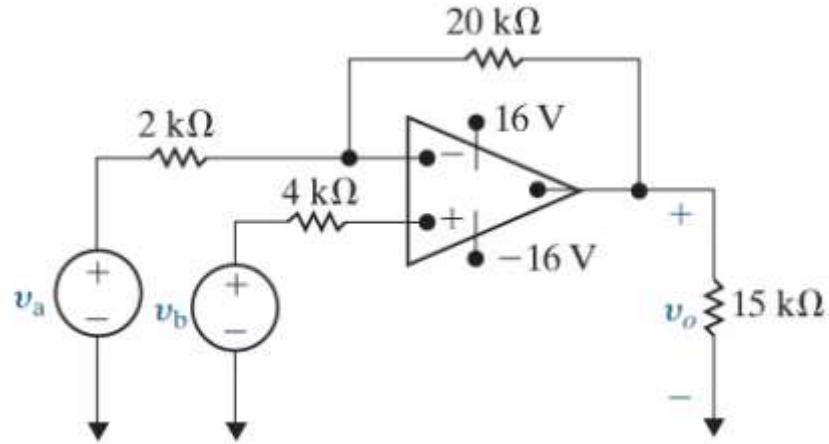
b. Calculate v_o if $v_a = -1$ V and $v_b = 0$ V.

c. Calculate v_o if $v_a = 3$ V and $v_b = 3.5$ V.

d. Calculate v_o if $v_a = -2$ V and $v_b = 1$ V.

e. Calculate v_o if $v_a = 4$ V and $v_b = 0$ V.

f. If $v_b = 3$ V, specify the range of v_a such that the amplifier does not saturate.



a. Calculate v_o if $v_a = 2$ V and $v_b = 1$ V.

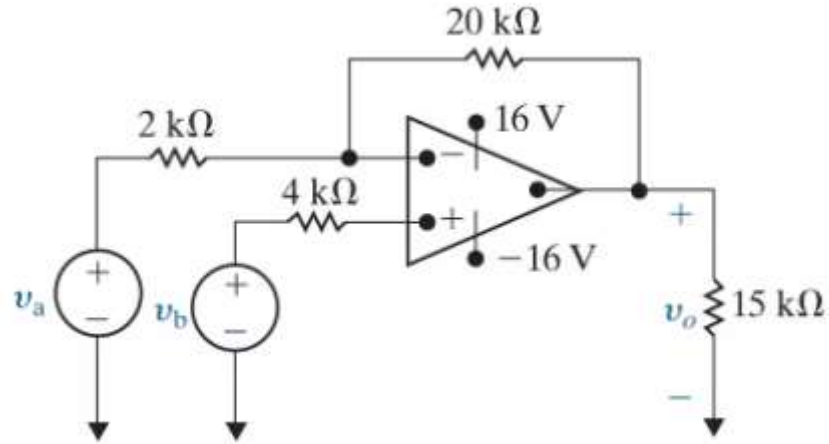
b. Calculate v_o if $v_a = -1$ V and $v_b = 0$ V.

c. Calculate v_o if $v_a = 3$ V and $v_b = 3.5$ V.

d. Calculate v_o if $v_a = -2$ V and $v_b = 1$ V.

e. Calculate v_o if $v_a = 4$ V and $v_b = 0$ V.

f. If $v_b = 3$ V, specify the range of v_a such that the amplifier does not saturate.



$$\frac{v_b - v_a}{2000} + \frac{v_b - v_o}{20{,}000} = 0, \text{ therefore } v_o = 11v_b - 10v_a.$$

[a] $v_a = 2$ V, $v_b = 1$ V, $\therefore v_o = -9$ V

[b] $v_a = -1$ V, $v_b = 0$ V, $\therefore v_o = 10$ V

[c] $v_a = 3$ V, $v_b = 3.5$ V, $\therefore v_o = 8.5$ V

[d] $v_a = -2$ V, $v_b = 1$ V, $\therefore v_o = 31$ V, saturated at 16 V

[e] $v_a = 4$V, $v_b = 0$ V, $\therefore v_o = -40$ V, saturated at -16 V

[f] If $v_b = 3$ V, $v_o = 33 - 10\ v_a = \pm 16$ V;

$\therefore$ v_o does not saturate if $1.7\text{ V} \leqslant v_a \leqslant 4.9$ V.

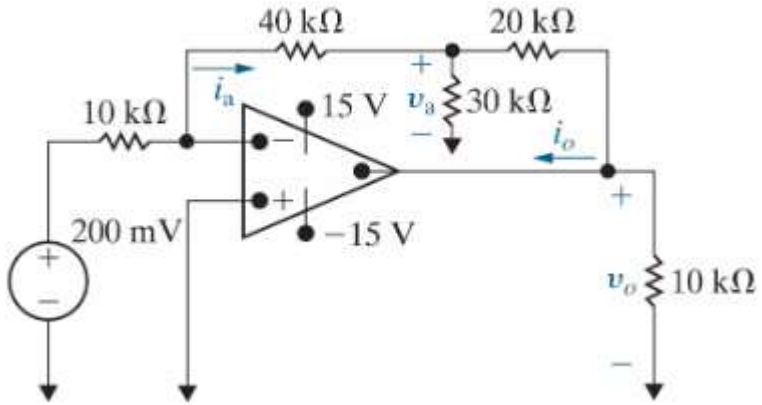
The op amp in the circuit in Fig. P5.6 is ideal. Calculate the following:

a. i_a

b. v_a

c. v_o

d. i_o



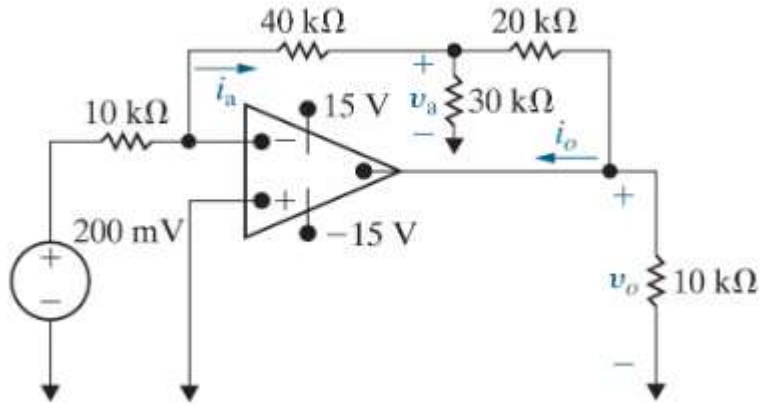
The op amp in the circuit in Fig. P5.6 is ideal. Calculate the following:

a. i_a

b. v_a

c. v_o

d. i_o



[a] $$i_a = \frac{200 \times 10^{-3}}{10{,}000} = 20\ \mu\text{A}$$

[b] $$\frac{0 - v_a}{40{,}000} = i_a \quad \text{so} \quad v_a = -40{,}000\ i_a$$

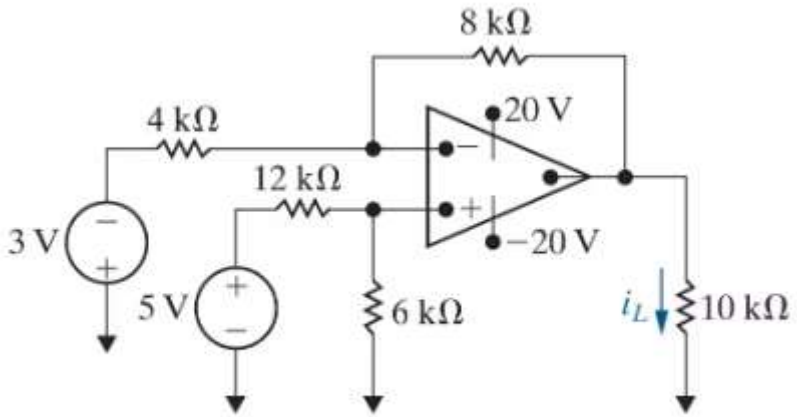
$$\therefore\ v_a = -0.8\ \text{V}$$

[c] $$\frac{v_a}{40{,}000} + \frac{v_a}{30{,}000} + \frac{v_a - v_o}{20{,}000} = 0;$$

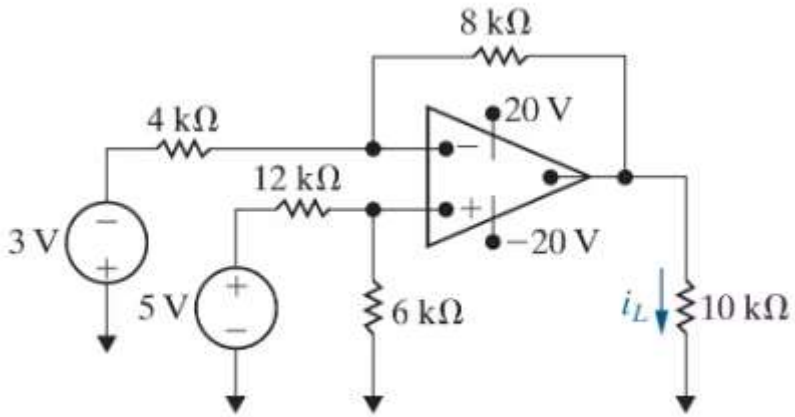
$$\therefore\ v_o = 2.167\ v_a = -1.73\ \text{V}$$

[d] $$i_o = \frac{-v_o}{10{,}000} + \frac{v_a - v_o}{20{,}000} = 219.5\ \mu\text{A}.$$

Find i_L (in milliamperes) in the circuit in Fig. P5.7.



Find i_L (in milliamperes) in the circuit in Fig. P5.7.



$$v_p = \frac{6000}{6000 + 12{,}000}(5) = 1.667 \text{ V} = v_n$$

$$\frac{v_n + 3}{4000} + \frac{v_n - v_o}{8000} = 0$$

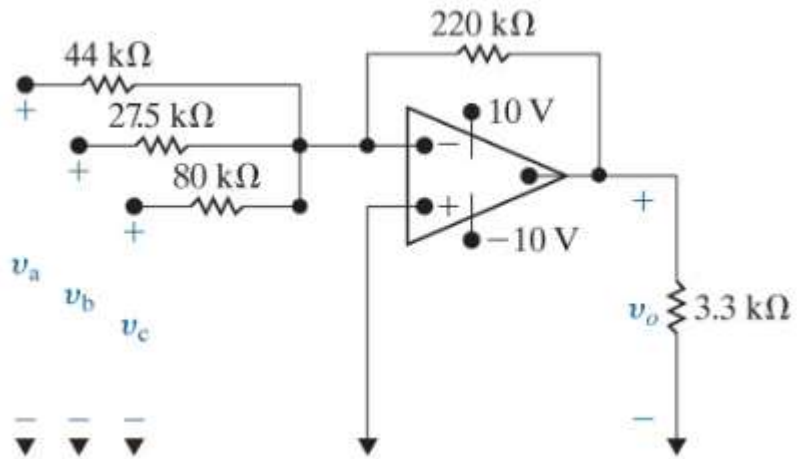
$$v_o = 6 + 3v_n = 11 \text{ V}$$

$$i_L = \frac{v_o}{10{,}000} = \frac{11}{10{,}000} = 1.1 \text{ mA}$$

a. What circuit configuration is shown in this figure?

b. Find v_o if $v_a = 1$ V, $v_b = 1.5$ V, and $v_c = -4$ V.

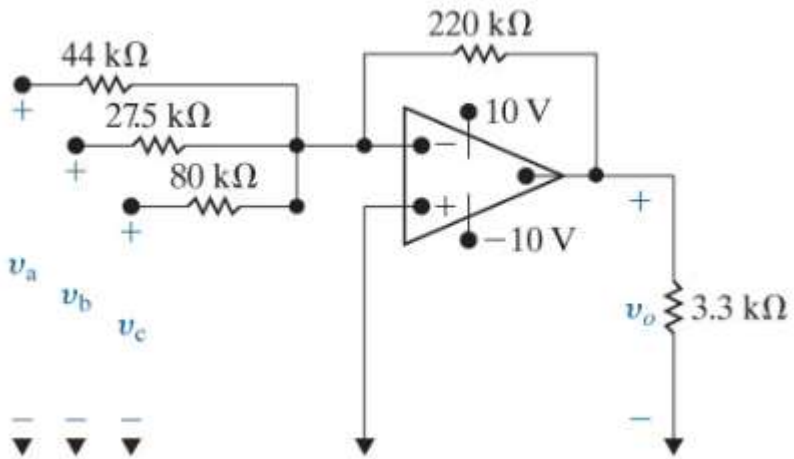
c. The voltages v_a and v_c remain at 1 V and -4 V, respectively. What are the limits on v_b if the op amp operates within its linear region?



a. What circuit configuration is shown in this figure?

b. Find v_o if $v_a = 1$ V, $v_b = 1.5$ V, and $v_c = -4$ V.

c. The voltages v_a and v_c remain at 1 V and -4 V, respectively. What are the limits on v_b if the op amp operates within its linear region?



[a] This circuit is an example of an inverting summing amplifier.

[b] $$v_o = -\frac{220}{44}v_a - \frac{220}{27.5}v_b - \frac{220}{80}v_c = -5 - 12 + 11 = -6 \text{ V}$$

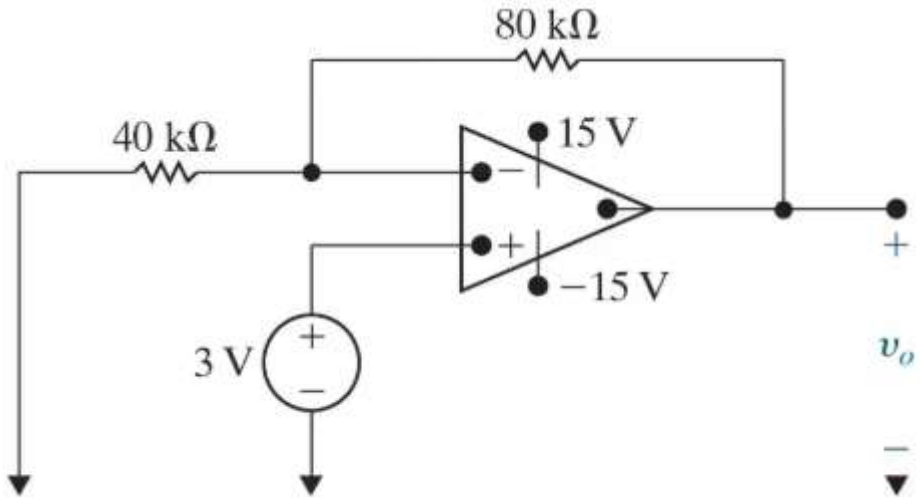
[c] $v_o = -6 - 8v_b = \pm 10$

$\therefore\ v_b = -0.5$ V when $v_o = 10$ V;

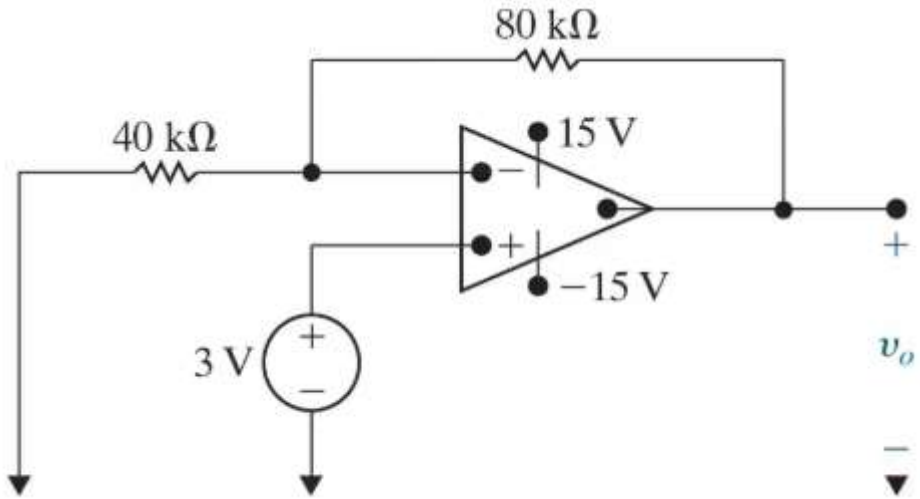
$v_b = 2$ V when $v_o = -10$ V

$\therefore\ -0.5 \text{ V} \leqslant v_b \leqslant 2 \text{ V}$

Calculate v_o.



Calculate v_o.



The circuit shown is a non-inverting amplifier.

We assume the op amp to be ideal, so $v_n = v_p = 3$ V. Write a KCL equation at v_n:

$$\frac{3}{40{,}000} + \frac{3 - v_o}{80{,}000} = 0.$$

Solving,

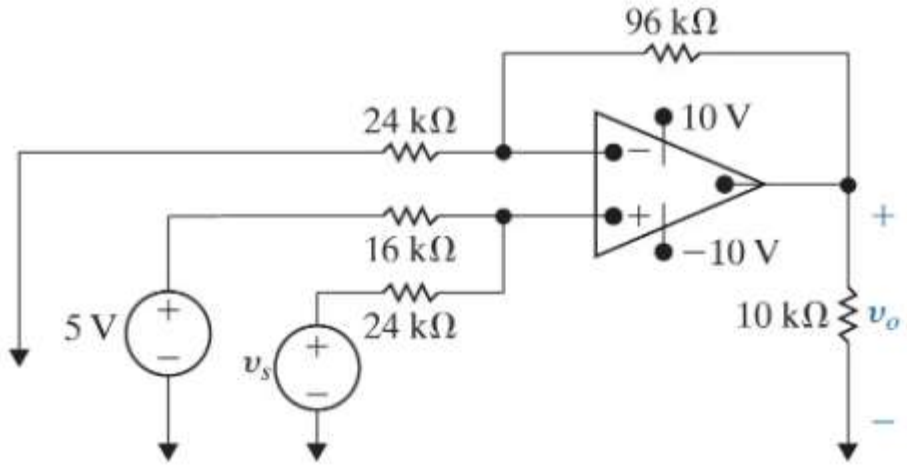
$v_o = 9$ V.

The op amp in the circuit of Fig. P5.23 is ideal.

a. What op amp circuit configuration is this?

b. Find v_o in terms of v_s.

c. Find the range of values for v_s such that v_o does not saturate and the op amp remains in its linear region of operation.

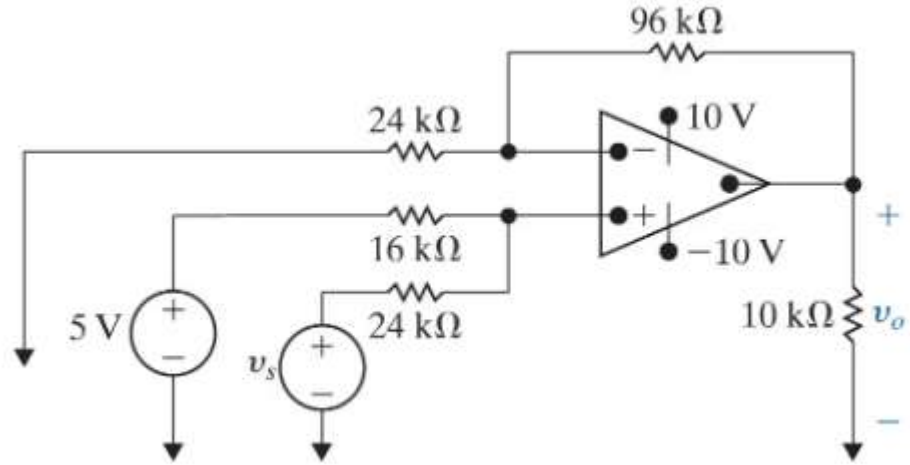


The op amp in the circuit of Fig. P5.23 is ideal.

a. What op amp circuit configuration is this?

b. Find v_o in terms of v_s.

c. Find the range of values for v_s such that v_o does not saturate and the op amp remain in its linear region of operation.



[a] This circuit is an example of a non-inverting summing amplifier.

[b] Write a KCL equation at v_p and solve for v_p in terms of v_s:

$$\frac{v_p - 5}{16{,}000} + \frac{v_p - v_s}{24{,}000} = 0$$

$$3v_p - 15 + 2v_p - 2v_s = 0 \quad \text{so} \quad v_p = 2v_s/5 + 3$$

Now write a KCL equation at v_n and solve for v_o:

$$\frac{v_n}{24{,}000} + \frac{v_n - v_o}{96{,}000} = 0 \quad \text{so} \quad v_o = 5v_n$$

Since we assume the op amp is ideal, $v_n = v_p$. Thus,

$$v_o = 5\,(2v_s/5 + 3) = 2v_s + 15$$

[c] $2v_s + 15 = 10 \quad \text{so} \quad v_s = -2.5$ V

$2v_s + 15 = -10 \quad \text{so} \quad v_s = -12.5$ V

Thus, $\quad -12.5 \text{ V} \leqslant v_s \leqslant -2.5 \text{ V}.$